Ternary modules on the flags of the plane of order four

Private derivation and exact certificate: independent review pending

This bounded note treats L050, Theorem 1(a), Proposition 1, Corollary 1 and Remarks 1–2. The source is Leemans–Rodrigues, *Linear codes with complementary duals from some strongly regular subgraphs of the McLaughlin graph*, Math. Commun. 21 (2016), 239–249. We reuse the common matrix-algebra, minimum-support and transitive information-set arguments. Complete previously accepted enumerations are retained for the two main codes. The two additional codes in Remark 2 require only the bounded search proved below.

1 The graph and the two principal codes

Use $\mathbb{F}_4 = \mathbb{F}_2[z]/(z^2 + z + 1)$ and the 105 incident point–line flags of $\mathrm{PG}(2, 4)$. Join distinct flags $(p, \ell), (q, m)$ when $p = q$, or $\ell = m$, or neither cross incidence $p \in m$, $q \in \ell$ holds. The complete field construction checks parameters $(105, 72, 51, 45)$.

The printed adjacency sentence on page 242 does not state this condition correctly. Reading its clauses conjunctively gives valency eight; reading them all disjunctively gives valency forty. Our identification with the intended graph therefore retains the uniqueness of the strongly regular graph with parameters $(105, 72, 51, 45)$, obtained from uniqueness of its complement in Coolsaet's Theorem 3, cited as source reference [6]. The parameter check alone would not identify an arbitrary graph with a named source graph without this input.

Let A be the adjacency matrix over $F = \mathbb{F}_3$, and write

$$C = \langle \text{rows of } A \rangle, \qquad D = \langle A_i - A_1 : 1 \le i \le 105 \rangle.$$

The parameter identity gives $A^2 = 0$, so C and D are self-orthogonal. Exact row reduction gives $\dim C = 20$, $\dim D = 19$, and

$$C = D \oplus Fj, \tag{1}$$

where j is the constant vector. Both bases agree byte for byte with the separately completed enumeration. The two complete distributions give

	dim	d	minimum words	$(\dim C^\perp, d^\perp, \text{minimum words})$
C	20	33	770	$(85, 6, 15680)$
D	19	36	3360	$(86, 5, 84)$.

The complete dual distributions are checked again by the classical MacWilliams identity. No 3^{20} or 3^{19} enumeration is repeated. The retained distributions agree entry by entry with the two printed tables.

2 The full group and irreducibility

The literal coordinate permutations in the certificate preserve every entry of A. Their closure has order 241920, and their action has rank four. They generate a group G with a normal subgroup $N \cong \mathrm{PSL}_3(4)$ of order 20160 and an explicit complement $Q \cong D_{12}$ of order twelve. The complement consists of a diagonal projectivity, field conjugation and point–line duality.

The checker derives their actual flag permutations, verifies $N \cap Q = 1$ and $NQ = G$, and checks the group identifications. Thus $G \cong L_3(4) : D_{12}$, with this action fixed.

The action on the row-reduced 19-dimensional basis of D is checked directly. The generator matrices span a matrix algebra of dimension $19^2 = 361$: the verifier evaluates independent generator words and row-reduces their flattened matrices. A nonzero invariant subspace stable under all these matrices is stable under every endomorphism, hence equals D. Independence persists over field extensions, so D is absolutely irreducible. Consequently (1) is an internal decomposition into two simple modules, as required in Proposition 1.

The complete minimum-word set of C is obtained as four disjoint orbits of literal weight-33 words, of lengths $105, 105, 280, 280$. The accepted count 770 proves exhaustion, and also recovers Remark 1. There are 385 distinct supports. As for the 120- and 176-coordinate codes, form their bipartite coordinate–support incidence graph. Refine vertex colours by old colour and neighbour-colour multiplicities, keeping the two parts initially distinct. Successively individualizing coordinates $1, 2, 3, 4, 7, 12, 6$ gives cell sizes

$$105, 8, 3, 2, 4, 3, 4.$$

All coordinate colours then differ. At each step a stabilizer orbit is contained in the selected colour class; after the final step the stabilizer is trivial. Thus

$$|\operatorname{Aut}(C)| \le 105 \cdot 8 \cdot 3 \cdot 2 \cdot 4 \cdot 3 \cdot 4 = 241920.$$

Since $G \le \operatorname{Aut}(\Gamma) \le \operatorname{Aut}(C)$, equality follows. Every automorphism of D fixes j and hence preserves $D + Fj = C$. Conversely G preserves D, so

$$\operatorname{Aut}(D) = \operatorname{Aut}(C) = \operatorname{Aut}(\Gamma) = L_3(4) : D_{12}. \tag{2}$$

These are coordinate-permutation groups, not monomial groups.

Both retained distributions have exactly 3360 words of weight 36. Since $D \subset C$, all such words of C are in D. Their span is a nonzero G-submodule of the irreducible D, and is therefore D. This establishes the source's intrinsic definition of its codimension-one code, rather than identifying it only by parameters.

3 Uniqueness inside the permutation module

Put $M = F^{105}$. A supplied basis gives invariant prefixes of dimensions

$$0, 1, 20, 39, 84, 85, 104, 105.$$

The factor dimensions are $1, 19, 19, 45, 1, 19, 1$. All factors are simple. For the factors of dimensions one and nineteen, this is verified by the full matrix algebra as above. To avoid an unnecessarily large elimination for the 45-dimensional factor, the certificate uses the following equivalent sufficient test.

Lemma 1. *Let $\mathcal{A}$ be an algebra generated by matrices on F^d. Suppose it contains a rank-one matrix $P = vu$, and that the vectors gv and uh, with g, h words in its generators, each span their respective column and row spaces of dimension d. Then $\mathcal{A} = M_d(F)$.*

Proof. Choose d such columns and d such rows forming bases. Their d^2 outer products are a basis of $M_d(F)$; each is $gPh \in \mathcal{A}$. □

Here a recorded polynomial of a recorded sum of generator words gives P of rank one. The checker evaluates it and produces both bases of size 45. Hence its algebra has dimension 2025,

proving simplicity over F and after extension. This is a certificate of the concrete factor, not an assumed classification of representations.

For each of the three 19-dimensional factors S_i, all linear equations

$$XP_g = (S_i)_g X$$

are solved, where P_g is the actual 105-coordinate permutation matrix. The full solution-space dimensions are $0, 1, 1$ in the displayed filtration order. Every basis solution has all rows in D, hence every homomorphism $S_i \to M$ has image in D.

Any 19-dimensional submodule of M is simple. Indeed its composition factors occur among those of M; dimension 45 is impossible, while the three available one-dimensional factors alone cannot account for dimension nineteen. Its sole factor must therefore have dimension nineteen. Such a simple submodule is isomorphic to one of the S_i, and its inclusion is the image of a corresponding homomorphism into M. The preceding complete equations force its image to be D. Thus D is the unique submodule of dimension nineteen in this permutation representation, proving the first assertion of Remark 2.

4 The other two twenty-dimensional modules

The certificate supplies invariant row-reduced bases for C_1, C_2, besides $C_3 = C$. Their containments are

	$D \subseteq C_i$	$j \in C_i$
C_1	no	yes
C_2	yes	no
C_3	yes	yes.

The only one-dimensional submodule of M is Fj. To see this directly, an invariant line has a spanning vector with entries in $\{0, a, -a\}$. Transitivity makes its zero set empty. A nontrivial scalar action would interchange the two nonzero values, forcing their multiplicities to be equal, contrary to the odd length 105. Thus the line is fixed and constant. Every 20-dimensional submodule has factors of dimensions 19 and one, by the full composition series above. The containments and uniqueness just proved give

$$\dim \mathrm{Soc}(C_1) = 1, \quad \dim \mathrm{Soc}(C_2) = 19, \quad \dim \mathrm{Soc}(C_3) = 20.$$

The three modules are therefore reducible and pairwise nonisomorphic as FG-modules, not just inequivalent as codes.

For their remaining code parameters, we reuse the already proved transitive information-set argument. If a transitive group preserves a length-n, dimension-k code and I is an information set, averaging over the group shows that a word of weight w has a translate whose restriction to I has weight at most $\lfloor kw/n \rfloor$. For $(n, k) = (105, 20)$ and candidate distances $30, 28$, both the smaller-word and minimum-word tests require only messages of weight at most five. There are

$$\sum_{i=0}^{5} \binom{20}{i} 2^i = 583569$$

such ternary messages, including zero. The complete shared enumerator checks them once each. Its smallest nonzero output weights are $30, 28$, and literal witnesses attain those values. The averaging theorem therefore proves those minimum distances.

The unions of a literal witness orbit and its negative have sizes 672 and 720. They are invariant sets of minimum words. Exactly 342 and 396 of these words, respectively, have message weight at most five; these equal the complete restricted enumeration counts at the minimum weight. Thus every restricted minimum word is in the proposed invariant set.

Averaging an arbitrary minimum word into that restriction proves that the sets are complete. This recovers $C_1 = [105, 20, 30]_3$ with 672 minimum words and $C_2 = [105, 20, 28]_3$ with 720 minimum words, as printed in Remark 2. It neither assumes a complete 3^{20} enumeration nor asserts that a short search is the full weight distribution.

5 Qualifications and attribution

The source's proof of Corollary 1 does not justify its global statement that nineteen is the smallest dimension of a nontrivial irreducible F_3G-module. Literally that statement is false: G has a normal subgroup of index two, so its quotient acts nontrivially by the scalars $1, -1$ on a one-dimensional F_3-space. The certificate exhibits the index-two subgroup by adjoining the diagonal and field automorphisms to N; its order is 120960. No faithfulness hypothesis is stated in the printed claim. We prove the precise local replacement: the unique nonconstant simple submodule of dimension nineteen in this permutation module is D, and no nonconstant simple submodule has smaller dimension. No classification of all faithful irreducible representations is inferred.

The common refinement and matrix-algebra arguments are the same as in the earlier two code supplements. The averaging theorem and complete sparse-message enumerator are reused from the distinctly reviewed transitive-distance packet. The exact incidence geometry is classical, and the uniqueness input identifying the source graph remains explicitly retained. Source comparisons to historical optimal-code tables remain historical assertions. Discovery used GRAPE and MeatAxe; the final certificate checks literal generators, a full invariant filtration, matrix products and complete linear equations, without requiring their discovery outputs to be trusted. No whole-source admission, new source credit, historical priority, numerical compression or end-to-end Lean claim is made in this bounded packet.